Supplemental Information

S1 Supplemental Methods

S1.1 Transcriptomics data acquisition and processing

Brain-wide transcriptomics data are sourced from the Allen human brain atlas (AHBA, human.brain-map.org), including over 58k probes sampled across 3,702 brain locations from six donors. We use the abagen toolbox[1] for data preprocessing and employ the AAL atlas to map brain locations to ROIs, resulting in a number of 15,633 genes expressed across 116 ROIs. To ensure that the transcriptomic knowledge closely aligns with the underlying mechanisms of the corresponding brain diseases, we further filter the genes to focus on those associated with disease risk. Specifically, we utilize two large-scale meta-GWAS results: one from AD (21,982 AD cases, 41,944 controls) from the International Genomics of Alzheimer's Project (IGAP)[2] and another for ADHD[3] (20,183 ADHD cases, 35,191 controls). We use the MAGMA[4] to derive gene-level p-values from SNP-level meta-GWAS results and keep nominally significant ones (i.e., $p <$ 0.05). This process yields a total of 1,216 genes expressed across 116 ROIs for AD and 1,524 genes expressed across the same 116 ROIs for ADHD. These expression datasets are used to construct the corresponding T-RRI edge matrices for the AD and ADHD prediction tasks, respectively.

S1.2 Imaging data acquisition and processing

We utilized two brain disorder datasets in our study two brain disease cohorts (ADNI and ADHD-200) involving four imaging modalities (AV45-PET, FDG-PET, VBM-sMRI, and fMRI) to evaluate the performance of CAMM. The acquisition and preprocessing of each modality are described as follows.

S1.2.1 AV45-PET modality

The preprocessed [18F]Florbetapir PET (AV45) scans were obtained from the LONI database (adni.loni.usc.edu). Prior to download, all PET images underwent standard preprocessing procedures as described by Jagust et al.[5] and Yan et al.[6], including averaging, alignment to a standard anatomical space, resampling to a common voxel grid, smoothing to achieve a uniform resolution, and normalization to a cerebellar gray matter reference region, resulting in standardized uptake value ratio (SUVR) images. After download, each PET image was rigidly aligned to the corresponding T1-weighted MRI scan acquired at the same visit. The images were then spatially normalized to the Montreal Neurological Institute (MNI) space using transformation parameters derived from MRI segmentation, with a final voxel resolution of 2 × 2 × 2 mm. Regional amyloid burden values were subsequently extracted at the region-of-interest (ROI) level using the Automated Anatomical Labeling (AAL) atlas implemented in MarsBaR[7].

S1.2.2 FDG-PET modality

The preprocessed [18F]FDG-PET scans were obtained from the LONI database (adni.loni.usc.edu). After download, each scan was aligned to the corresponding T1-weighted MRI acquired at the same visit for each participant. The aligned PET images were then spatially normalized to the MNI space using transformation parameters from the MRI segmentation, with a final voxel resolution of 2 × 2 × 2 mm. Measurements of glucose metabolism were subsequently extracted at the ROI level using the AAL atlas implemented in MarsBaR[7]. More details can be found at Yao et al.[8].

S1.2.3 VBM-sMRI modality

To investigate brain-associated MRI imaging phenotypes, we analyzed data from the ADNI database, including MRI scans from participants[9]. Each participant underwent at least two baseline MP-RAGE scans at 1.5 T, following the ADNI MRI protocol[10]. The scans were processed using voxel-based morphometry (VBM), a widely employed automated MRI analysis method[11]. VBM analysis was conducted with SPM12, which generated unmodulated, normalized grey matter (GM) density maps with a voxel resolution of 1 × 1 × 1 mm, smoothed using a 10 mm full-width at half-maximum (FWHM) Gaussian kernel. For each participant, two independently processed maps were averaged to obtain a mean GM density map. Regional GM density values were extracted in the MNI space using the MarsBaR ROI toolbox[7]. Both ADNI dataset and ADHD-200 dataset were preprocessed with the same way.

S1.2.4 The fMRI modality

The preprocessed fMRI scans were obtained from the NITRC (`https://www.nitrc.org/`). The original fMRI data were preprocessed using a public toolbox named DPABI[12] (for Data Processing & Analysis of Brain Imaging, `http://rfmri.org/dpabi`). The preprocessing steps were as follows: (1) remove the first 10 volumes to ensure that the BOLD signal was stable; (2) slice timing, correct the difference due to acquisition times between slices in the volume; (3) head motion correction; (4) normalization, register the data to the EPI standard template and resample it to $3.0 \times 3.0 \times 3.0$ mm; and (5) spatial smoothing with a 6-mm full width at half maximum (FWHM) Gaussian kernel. Subjects whose head movement exceeded 2.0 mm were excluded. For the fMRI modality, which includes a temporal dimension, the dynamic imaging data were represented as a three-dimensional tensor $\mathbf{D}^m_{dynamic} = \{\mathbf{X}^m_1, \cdots, \mathbf{X}^m_n\} \in \mathbb{R}^{n \times p \times d}$, where each sample-specific matrix $\mathbf{X}^m_s \in \mathbb{R}^{p \times d}$ corresponds to p time points across d regions of interest (ROIs). We constructed sample-specific region-wise region-to-region interaction (R-RRI) networks for each individual by computing the Pearson correlation coefficient (PCC) between all pairs of ROIs over time. The resulting correlation matrices were thresholded using a modality-specific hyperparameter λ^m_r to generate a set of binary edge matrices $\mathcal{E}^m = \{\mathbf{E}^m_s\}^n_{s=1}$. SIP et al.[13] demonstrated that underlying dynamics in resting-state fMRI can be effectively characterized as noisy fluctuations around a single fixed point, supporting the utility of simplified temporal representations. In addition, Lee et al.[14] evaluated the discriminative power of features generated by principal component analysis (PCA) in distinguishing individuals with psychophysiological insomnia (PI) from healthy controls (HC), and found that PCA-based features achieved the best classification performance across all evaluated metrics. Motivated by these findings, we applied PCA to the temporal signals of each ROI and retained the first principal component to obtain a single representative value, which was used as the node feature in the R-RRI graph.

Table S1: Model performance under different PCA component selections

PCA component selection	NC vs. ADHD			
	ACC (%)	F1 (%)	AUC (%)	time (s/epoch)
1	80.2±1.3	79.3±1.4	80.1±1.8	0.54
10	74.1±1.3	72.7±1.6	78.7±1.1	1.08
20	77.5±1.5	76.3±1.6	75.7±1.3	1.4
50	75.8±2.1	75.8±1.8	72.2±1.9	1.61

To empirically validate the rationality of using PCA, we compared classification performance under different numbers of retained components (1, 10, 20, and 50). As shown in Table S1, the model achieves the highest performance when using only the first principal component (ACC: 80.2% ± 1.3, F1: 79.3% ± 1.4, AUC: 80.1% ± 1.8), while increasing the number of components leads to a gradual performance drop. Specifically, retaining 10 components yields slightly degraded results (ACC: 74.1%), and performance also declines with 20 or 50 components. These results indicate that the first principal component captures most of the discriminative variance for the NC vs. ADHD classification task. Moreover, reducing to a single component significantly accelerates training (0.54 s/epoch), demonstrating the efficiency and effectiveness of low-rank representation in fMRI data modeling. Based on these findings, we adopt the single-component compression strategy in the main experiments.

S1.3 Construction of single-sample level pattern

In this study, we adopted the SWEET method[15] to build individualized single-sample networks, as illustrated in Figure S1, for each subject S_i, utilizing the feature (e.g., imaging) expression profiles derived from a cohort of n case samples. To address the issue of biased edge numbers introduced by inherent subpopulation structures, SWEET first computes the Pearson correlation coefficient (PCC) between every pair of samples based on their expression profiles across m features. This results in a sample-to-sample PCC matrix (Figure S1A).

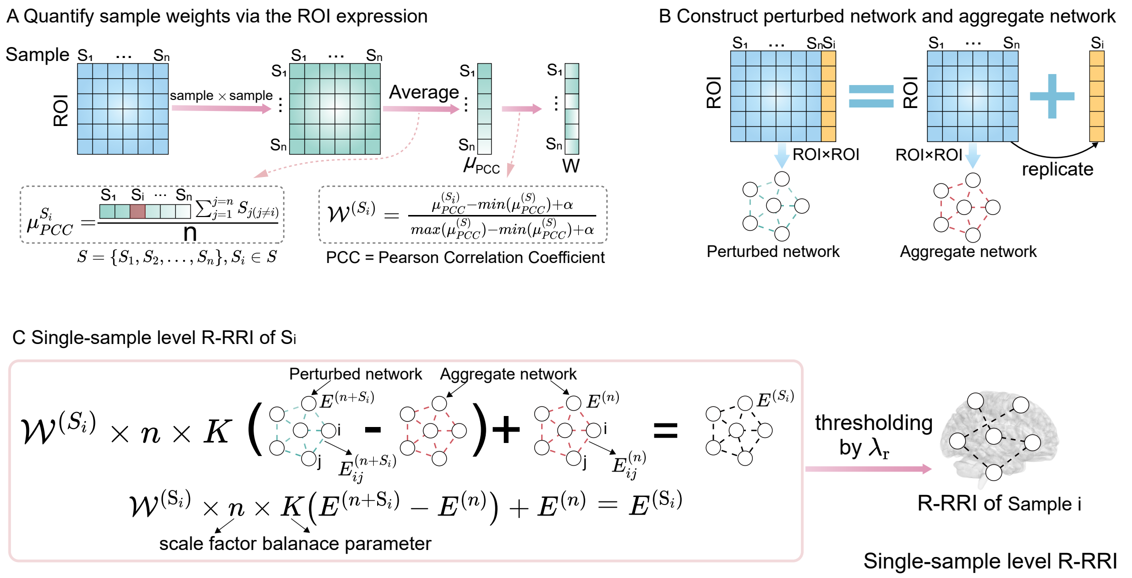

Figure S1: Illustration of single-sample level pattern construction. (A) Quantification of sample weights to obtain the expression matrix sample weight $W^{(S_i)}$ for each sample S_i. (B) Construction of the aggregate network and the perturbed network based on expression matrices. The aggregate network is established by calculating the PCC between the expression levels of any two features. A copy of each sample S_i is added to the overall samples to calculate the PCC of every feature pair and build a perturbed network. (C) The single-sample network edge $E^{(S_i)}$ for each sample S_i is inferred by evaluating the difference between the aggregate and perturbed networks, integrated with the corresponding sample weights, and then thresholded using λ_r.

For each sample S_i, the average PCC value, denoted as v^{S_i}, is calculated and subsequently used to derive a sample-specific weight $W^{(S_i)}$ that reflects its similarity to the overall cohort. The sample weight is formulated as:

$$W^{(S_i)} = \frac{\mu_{\text{PCC}}^{(S_i)} - \min\left(\mu_{\text{PCC}}^{(S)}\right) + \alpha}{\max\left(\mu_{\text{PCC}}^{(S)}\right) - \min\left(\mu_{\text{PCC}}^{(S)}\right) + \alpha}, \tag{1}$$

where $\mu_{\text{PCC}}^{(S_i)}$ represents the average PCC between sample S_i and all other $n-1$ samples, $\mu_{\text{PCC}}^{(S)}$ is the collection of mean PCCs from all samples in the cohort, and α is a small constant (e.g., 0.01) added to the denominator to avoid division by zero.

Next, the PCC between features i and j, denoted as $E_{ij}^{(n)}$, is calculated across all n samples to form the aggregate network structure (Figure S1B and C). To derive a perturbed version of the network for a specific sample S_i, its expression profile is duplicated and added to the original sample group, after which the PCC is recomputed over the $n+1$ samples. The resulting perturbed edge is denoted as $E_{ij}^{(n+S_i)}$. To personalize the network for each sample, we incorporate the sample-specific weight $W^{(S_i)}$ to adjust for subpopulation effects. The final edge score for the single-sample network of S_i, denoted by $E_{ij}^{(S_i)}$, is computed as:

$$E_{ij}^{(S_i)} = W^{(S_i)} \times n \times K\left(E_{ij}^{(n+S_i)} - E_{ij}^{(n)}\right) + E_{ij}^{(n)}, \tag{2}$$

where n represents the total number of case samples used to scale the correlation difference, and K is a predefined balance coefficient (empirically set to 10% of the total sample size) to control the contribution of the sample-specific perturbation.

According to the systematic evaluation reported in[15], setting the balance parameter K to 10% resulted in optimal median values for both the average degree exponent γ and the coefficient of determination R^2. A higher sample weight $W^{(S_i)}$ implies that sample S_i shares stronger similarity with the remaining cohort members, indicating potential membership in a larger subpopulation. Conversely, a lower weight may reflect distinctiveness, suggesting that the sample might belong to a rarer sub-cohort. This weighting strategy is designed to mitigate the influence of sample-level heterogeneity on differential correlation estimation. To determine the statistical significance of each edge $E_{ij}^{(S_i)}$, we conduct a z-score–based test:

$$z\left(E_{ij}^{(S_i)}\right) = \frac{E_{ij}^{(S_i)} - \mu}{\sigma}, \tag{3}$$

where μ and σ denote the overall mean and standard deviation of edge values computed across all single-sample networks. An edge is considered statistically significant in the context of sample S_i if its absolute z-score surpasses a critical threshold (e.g., 1.960 or 2.576, corresponding to $P \leq 0.05$ or 0.01).

Finally, in our framework, we obtain a collection of sample-specific edge matrices, denoted as $\mathcal{E}^m = \{\mathbf{E}_i^m\}_{i=1}^n$, by applying a sparsity-inducing threshold to each correlation-based network. This is achieved using a modality-dependent parameter λ_r^m, which facilitates interpretability while maintaining biologically relevant connections.

S2 Supplemental Results

S2.1 Hyperparameter analysis of thresholds for RRI construction

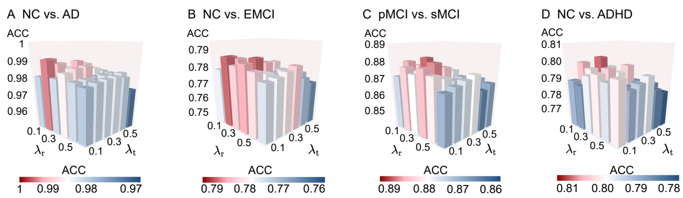

Figure S2: Comparison of hyperparameters for RRI thresholding.

We analyze the effect of varying threshold parameters used in constructing the T-RRI and R-RRI networks. In practice, λ_t and λ_r are tuned using grid search within the range $[0.1, 0.2, \cdots, 0.6]$. Figure S2 shows that the performance remains stable across different settings, indicating the robustness of our model. The slight variations in accuracy further indicate that both transcriptomics- and radiomics-derived RRIs contribute to the overall effectiveness of disease prediction.

S2.2 Sensitivity analysis of the softmax temperature and regularization strength

We conducted sensitivity analyses to evaluate the robustness of CAMM with respect to two key hyperparameters: the softmax temperature T and the regularization strength η_3 used in the confidence-guided regularization. As shown in Figure S4(A), CAMM exhibits stable classification performance across a wide range of temperature values for all four tasks. In general, moderate temperature values yield better performance, while both lower and higher temperatures lead to marginal accuracy reductions. Importantly, no abrupt performance degradation is observed, indicating that the proposed confidence calibration mechanism is not sensitive to the choice of T. Figure S4(B) illustrates the impact of the regularization strength η_3 on model performance. Similar to the temperature analysis, CAMM maintains consistent accuracy across a broad range of η_3 values. Moderate regularization strengths tend to achieve the best results, whereas excessively small or large values result in slight performance drops. Overall, these results demonstrate that CAMM is stable with respect to its key hyperparameters and does not rely on finely tuned parameter settings to achieve strong performance.

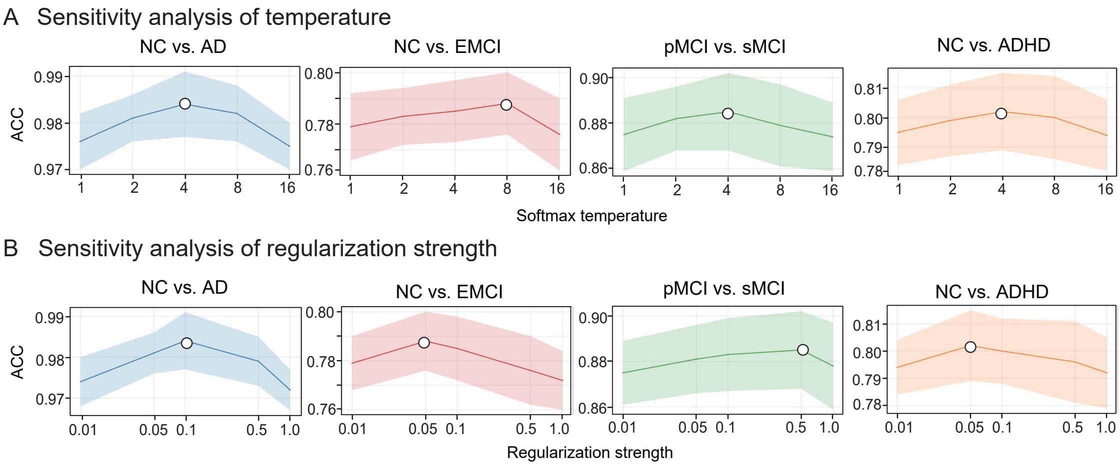

Figure S3: Sensitivity analysis of key hyperparameters in CAMM, including (A) the softmax temperature T and (B) the regularization strength η_3 used in the confidence-guided regularization. Classification accuracy (ACC) is reported, with mean and standard deviation (shaded area) are shown.

S2.3 Identified biomarkers and functional relevance

To standardize regional analysis, we adopted the Automated Anatomical Labeling (AAL) atlas to define regions of interest (ROIs). Specifically, all reported results in this study are mapped to the AAL brain regions, with each region identified by both its full anatomical name (AAL region) and its corresponding abbreviation. The mapping between region names and their abbreviations is summarized in Table S2.

S2.3.1 Co-functional interpretation of top ROIs

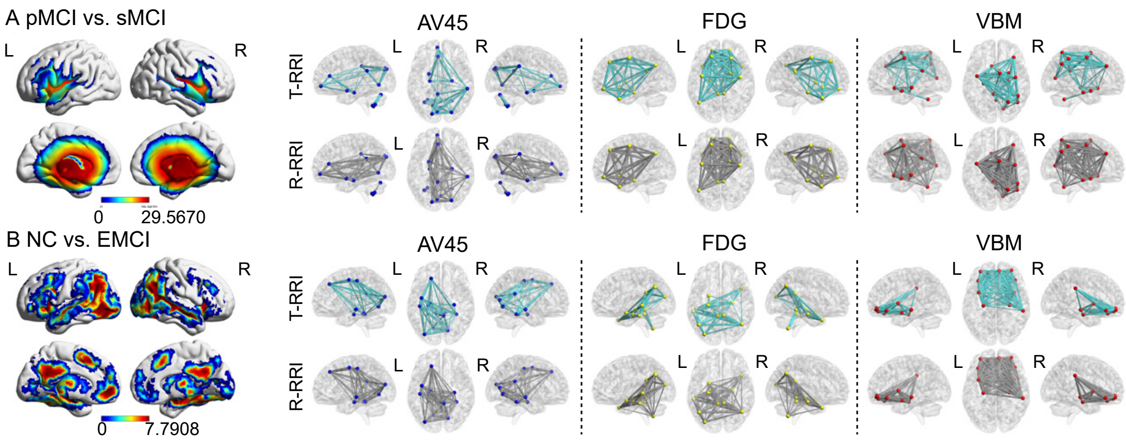

Figure S4: Coactivation maps for co-functional interpretation of top ROIs. (A) Coactivation maps and modality-specific connectivity (transcriptomics: top panel, gray edges; radiomics: bottom panel, green edges) of the top ROIs for the pMCI vs. sMCI task. (B) Corresponding results for the NC vs. EMCI task.

We next sought to dissect the biological underpinnings of the most discriminative ROIs identified in two representative classification tasks: pMCI vs. sMCI and NC vs. EMCI. As shown in Figure S4, transcriptomic coactivation patterns were examined alongside modality-specific connectivity networks derived from AV45, FDG, and VBM imaging.

For the pMCI vs. sMCI task (Figure S4A), transcriptomic coactivation prominently involved bilateral temporal and parietal cortices, regions that are central to early AD progression[16]. Radiomic networks displayed complementary features. AV45 highlighted cross-hemispheric posterior connections, reflecting amyloid-driven network fragility. FDG and VBM instead captured localized

Table S2: Abbreviation of the AAL brain regions

Index	AAL region	Abbreviation	Index	AAL region	Abbreviation
1	Precentral_L	PreCG-L	59	Parietal_Sup_L	SPG-L
2	Precentral_R	PreCG-R	60	Parietal_Sup_R	SPG-R
3	Frontal_Sup_L	SFGdor-L	61	Parietal_Inf_L	IPL-L
4	Frontal_Sup_R	SFGdor-R	62	Parietal_Inf_R	IPL-R
5	Frontal_Sup_Orb_L	ORBsup-L	63	SupraMarginal_L	SMG-L
6	Frontal_Sup_Orb_R	ORBsup-R	64	SupraMarginal_R	SMG-R
7	Frontal_Mid_L	MFG-L	65	Angular_L	ANG-L
8	Frontal_Mid_R	MFG-R	66	Angular_R	ANG-R
9	Frontal_Mid_Orb_L	ORBmid-L	67	Precuneus_L	PCUN-L
10	Frontal_Mid_Orb_R	ORBmid-R	68	Precuneus_R	PCUN-R
11	Frontal_Inf_Oper_L	IFGoperc-L	69	Paracentral_Lobule_L	PCL-L
12	Frontal_Inf_Oper_R	IFGoperc-R	70	Paracentral_Lobule_R	PCL-R
13	Frontal_Inf_Tri_L	IFGtriang-L	71	Caudate_L	CAU-L
14	Frontal_Inf_Tri_R	IFGtriang-R	72	Caudate_R	CAU-R
15	Frontal_Inf_Orb_L	ORBinf-L	73	Putamen_L	PUT-L
16	Frontal_Inf_Orb_R	ORBinf-R	74	Putamen_R	PUT-R
17	Rolandic_Oper_L	ROL-L	75	Pallidum_L	PAL-L
18	Rolandic_Oper_R	ROL-R	76	Pallidum_R	PAL-R
19	Supp_Motor_Area_L	SMA-L	77	Thalamus_L	THA-L
20	Supp_Motor_Area_R	SMA-R	78	Thalamus_R	THA-R
21	Olfactory_L	OLF-L	79	Heschl_L	HES-L
22	Olfactory_R	OLF-R	80	Heschl_R	HES-R
23	Frontal_Sup_Medial_L	SFGmed-L	81	Temporal_Sup_L	STG-L
24	Frontal_Sup_Medial_R	SFGmed-R	82	Temporal_Sup_R	STG-R
25	Frontal_Med_Orb_L	ORBmed-L	83	Temporal_Pole_Sup_L	TPOsup-L
26	Frontal_Med_Orb_R	ORBmed-R	84	Temporal_Pole_Sup_R	TPOsup-R
27	Rectus_L	REC-L	85	Temporal_Mid_L	MTG-L
28	Rectus_R	REC-R	86	Temporal_Mid_R	MTG-R
29	Insula_L	INS-L	87	Temporal_Pole_Mid_L	TPOmid-L
30	Insula_R	INS-R	88	Temporal_Pole_Mid_R	TPOmid-R
31	Cingulum_Ant_L	ACG-L	89	Temporal_Inf_L	ITG-L
32	Cingulum_Ant_R	ACG-R	90	Temporal_Inf_R	ITG-R
33	Cingulum_Mid_L	MCG-L	91	Cerebelum_Crus1_L	CERCRU1-L
34	Cingulum_Mid_R	MCG-R	92	Cerebelum_Crus1_R	CERCRU1-R
35	Cingulum_Post_L	PCG-L	93	Cerebelum_Crus2_L	CERCRU2-L
36	Cingulum_Post_R	PCG-R	94	Cerebelum_Crus2_R	CERCRU2-R
37	Hippocampus_L	HIP-L	95	Cerebelum_3_L	CER3-L
38	Hippocampus_R	HIP-R	96	Cerebelum_3_R	CER3-R
39	ParaHippocampal_L	PHG-L	97	Cerebelum_4_5_L	CER4_5-L
40	ParaHippocampal_R	PHG-R	98	Cerebelum_4_5_R	CER4_5-R
41	Amygdala_L	AMYG-L	99	Cerebelum_6_L	CER6-L
42	Amygdala_R	AMYG-R	100	Cerebelum_6_R	CER6-R
43	Calcarine_L	CAL-L	101	Cerebelum_7b_L	CER7b-L
44	Calcarine_R	CAL-R	102	Cerebelum_7b_R	CER7b-R
45	Cuneus_L	CUN-L	103	Cerebelum_8_L	CER8-L
46	Cuneus_R	CUN-R	104	Cerebelum_8_R	CER8-R
47	Lingual_L	LING-L	105	Cerebelum_9_L	CER9-L
48	Lingual_R	LING-R	106	Cerebelum_9_R	CER9-R
49	Occipital_Sup_L	SOG-L	107	Cerebelum_10_L	CER10-L
50	Occipital_Sup_R	SOG-R	108	Cerebelum_10_R	CER10-R
51	Occipital_Mid_L	MOG-L	109	Vermis_1_2	VER1_2
52	Occipital_Mid_R	MOG-R	110	Vermis_3	VER3
53	Occipital_Inf_L	IOG-L	111	Vermis_4_5	VER4_5
54	Occipital_Inf_R	IOG-R	112	Vermis_6	VER6
55	Fusiform_L	FFG-L	113	Vermis_7	VER7
56	Fusiform_R	FFG-R	114	Vermis_8	VER8
57	Postcentral_L	PoCG-L	115	Vermis_9	VER9
58	Postcentral_R	PoCG-R	116	Vermis_10	VER10

interactions in temporal and cingulate cortices, consistent with early hypometabolism and atrophy in progressive MCI. In the NC vs. EMCI comparison (Figure S4B), transcriptomic activity shifted toward frontal and medial temporal regions, a pattern aligned with early neurofunctional alterations in preclinical AD. AV45 again revealed broader and more integrated connectivity than FDG or VBM, suggesting heightened sensitivity to incipient pathological changes. Indeed, ten of the top fifteen ROIs prioritized for the NC vs. AD task were AV45-derived, underscoring its discriminative capacity across disease stages.

Importantly, transcriptomics-derived networks (T-RRI, gray edges) and radiomics-derived networks (R-RRI, green edges) revealed distinct yet complementary signatures. Transcriptomic patterns captured upstream molecular coexpression, whereas radiomics delineated spatially constrained and phenotype-specific interactions. This divergence highlights a coherent mechanistic hierarchy in which molecular coordination precedes and shapes downstream structural and metabolic phenotypes. These results demonstrate that our framework not only integrates heterogeneous modalities but also yields biologically interpretable and disease-consistent ROI-level insights. This supports its utility for biomarker discovery and stage-specific modeling in neurodegenerative disorders.

S2.3.2 Discriminative brain region signatures revealed by unimodal region-level ablation

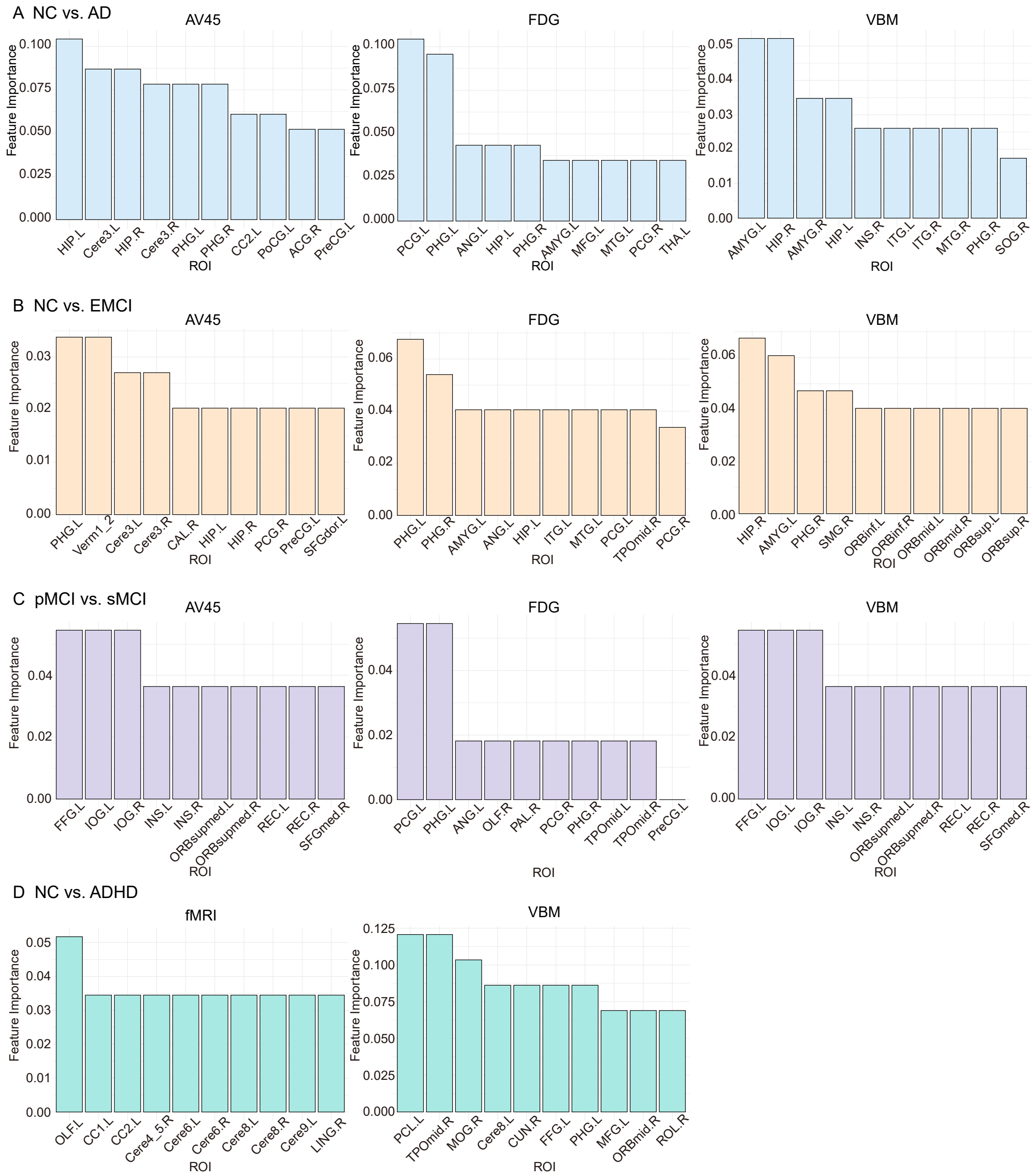

Figure S5: Region-level ablation analysis within individual imaging modalities for (a) NC vs. AD, (b) NC vs. EMCI, (c) pMCI vs.sMCI, and (d) NC vs. ADHD tasks.

To delineate the regional contributions of different brain areas across modalities, we conducted a systematic unimodal region-level ablation analysis over four representative classification tasks: NC vs. AD, NC vs. EMCI, pMCI vs. sMCI, and NC vs. ADHD (Figure S5). By selectively masking individual ROIs within each modality, we ranked their discriminative power according to the drop in classification performance, thereby revealing both modality-specific patterns and disease-relevant regional signatures.

In AD-related comparisons, canonical pathological hubs consistently emerged at the top of the rankings. The hippocampus (HIPP) showed the highest discriminative value in AV45 and remained among the leading regions in FDG and VBM, in agreement with extensive neuropathological and imaging literature that highlights its early and severe involvement in AD[17,18]. Hippocampal atrophy is tightly linked to disease progression, particularly in medial temporal subregions, while amyloid deposition and tau hyperphosphorylation first target CA1 and subsequently extend to CA2–CA4 and the dentate gyrus, driving synaptic loss and memory impairment. Complementary to the hippocampal findings, the parahippocampal gyrus (PHG) ranked highly across all three modalities, underscoring its role as an early target of tau pathology[19]. The posterior cingulate gyrus (PCG) was the most discriminative ROI in FDG, consistent with its role as a hub of the default mode network and its vulnerability to metabolic decline[20,21]. The amygdala (AMYG) emerged as the most important region in VBM and also ranked prominently in FDG, highlighting its structural and metabolic vulnerability in AD progression. The angular gyrus (ANG), identified in FDG, further supports its role in higher-order cognitive functions such as semantic processing and memory integration[22,23]. Together, these findings indicate that AV45 highlights hippocampal amyloid burden, FDG emphasizes metabolic impairment in temporolimbic and parietal hubs, and VBM captures prominent atrophy in medial temporal and limbic regions, particularly the amygdala.

In the NC vs. EMCI comparison (Figure S5B), region-level ablation highlighted a set of temporolimbic and association cortices that are consistent with early pathological alterations in preclinical AD. AV45 identified bilateral hippocampal (HIPP) and parahippocampal (PHG) regions among the top discriminative ROIs, but also revealed high importance of cerebellar vermis (Vermis) and lobules III–V (Cerebelum.3, Cerebelum.4-5), suggesting that amyloid-related network perturbations may extend beyond cortical hubs to subcortical and cerebellar structures at this stage. FDG predominantly emphasized the medial temporal lobe, with bilateral PHG, HIPP, and amygdala (AMYG) ranking highest, accompanied by ANG, middle temporal gyrus (MTG), and PCG, in line with early hypometabolism within the limbic–parietal axis. VBM, in contrast, highlighted a broader distribution of structural vulnerability: right hippocampus (HIPP.R) was the strongest driver, followed by ANG, supramarginal gyrus (SMG), calcarine cortex (CAL), and orbitofrontal cortices (ORBinf, ORBmid, ORBsup). This divergence across modalities underscores a mechanistic gradient in EMCI: molecular deposition captured by AV45 manifests in temporolimbic and cerebellar sites, metabolic decline detected by FDG is concentrated in medial temporal and associative cortices, while VBM reveals a more spatially distributed atrophic pattern spanning temporal, parietal, occipital, and frontal regions.

In earlier disease stages, additional cortical regions gained discriminative weight. The insula was especially important in pMCI vs. sMCI tasks. Prior work has shown selective insular atrophy in AD and paradoxical functional hyperactivation in individuals at risk, suggesting compensatory mechanisms under pathological stress[24,25]. Together, these results highlight a temporal gradient in AD progression, in which hippocampal–parahippocampal–cingulate networks dominate at advanced stages, whereas insular alterations emerge as early markers in prodromal phases.

Extending the ablation framework to ADHD revealed a distinct set of discriminative ROIs spanning olfactory, cerebellar, parietal, occipital, temporal, subcortical, and frontal regions, consistent with the distributed nature of ADHD neuropathophysiology (Figure S5D). In the fMRI modality, the olfactory cortex (OLF) emerged as the most important region, aligning with reports of impaired olfactory identification and orbitofrontal dysfunction in ADHD[26,27]. Multiple cerebellar lobules, particularly lobule 6 (Cerebelum.6) bilaterally, also ranked highly, reflecting the growing recognition of cerebellar contributions to attention and executive regulation. Strikingly, the cerebellum emerged as a key discriminative structure, with Cerebelum.6 consistently identified across modalities. Cerebellum.6 has been strongly associated with IQ in ADHD[28] and exhibits high control centrality in network dynamics[29], underscoring its central role in executive and cognitive regulation. In the VBM modality, the paracentral lobule (PCL) was the strongest driver, reinforcing its role in motor–sensory integration and attentional control, with altered connectivity to cortico-striatal circuits previously linked to ADHD symptoms[30,31]. The mid-temporal pole (TPOrMid) also featured prominently, consistent with its involvement in social attention and emotion processing[32,33], while the caudate nucleus (Caud) emerged as a key subcortical contributor, highlighting cortico-striatal imbalance as a hallmark of ADHD.

S2.3.3 Region-level pairwise ablation identifies critical intra-modality interactions

To further investigate the intra-modality inter-regional dependencies underlying brain disorders, we conducted pairwise ablation experiments within individual imaging modalities. Figure S6 visualizes the region-level interaction patterns for each task and modality using circular chord diagrams, where the strength and prevalence of connections reflect the importance of regional co-contributions to disease classification. These patterns provide valuable insight into the cooperative roles of brain regions beyond their individual effects.

In the NC vs. AD task, the hippocampus–parahippocampal gyrus pair consistently emerged as a dominant interaction across AV45, FDG, and VBM modalities. These medial temporal lobe structures are anatomically contiguous and functionally interdependent. Previous studies have documented their joint structural degeneration in early AD, where hippocampal and parahippocampal atrophy correlate strongly with MMSE decline and memory impairment[34,35]. Functionally, while the hippocampus is central to episodic and semantic encoding, the parahippocampal

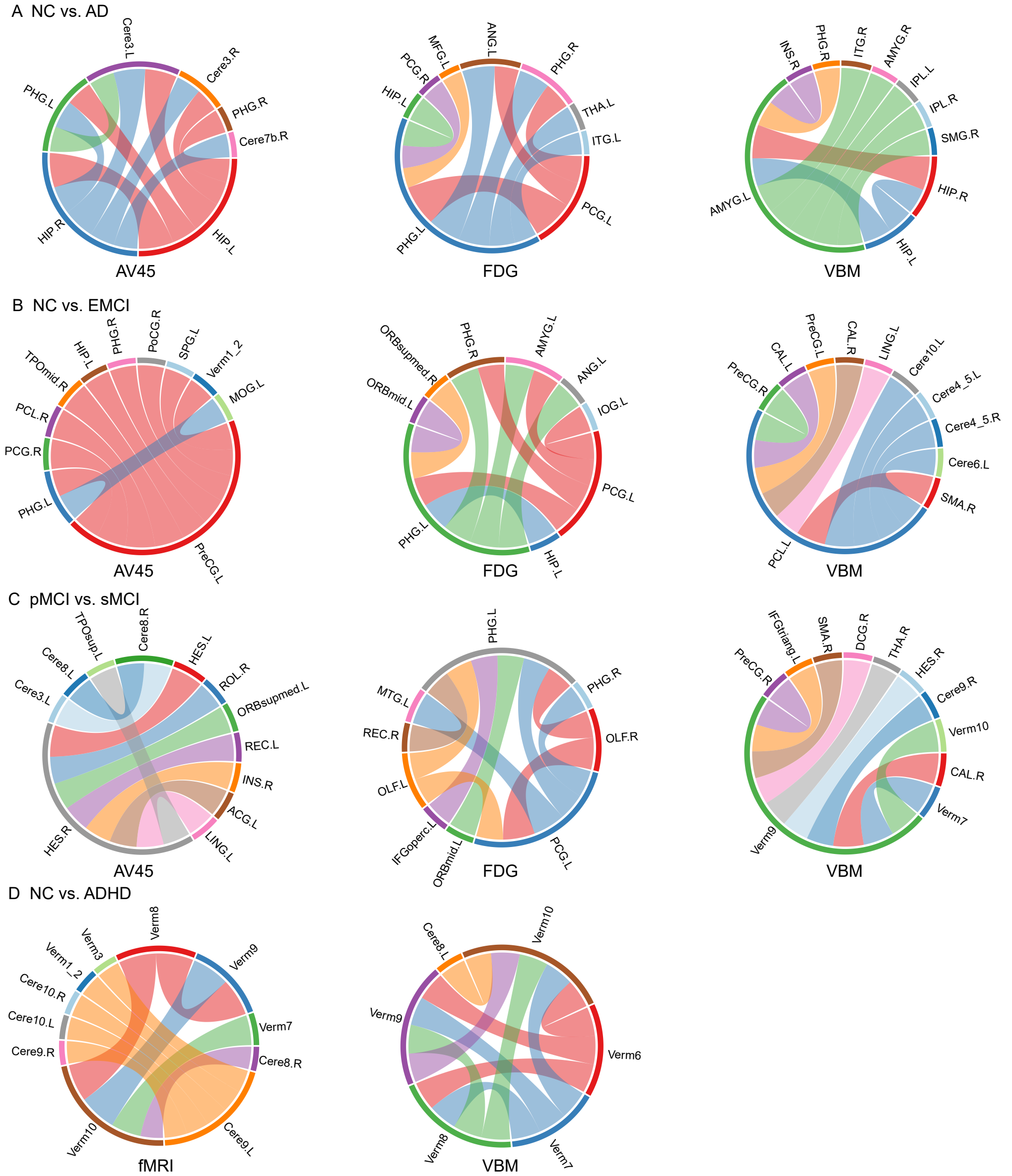

Figure S6: Region-level pairwise ablation analysis within individual modalities for intra-modality interaction patterns in (A) NC vs. AD, (B) NC vs. EMCI, (C) pMCI vs. sMCI, and (D) NC vs. ADHD tasks.

gyrus preferentially supports visuospatial memory. Their cooperative disruption thus reflects a breakdown of a complementary memory subsystem, providing a mechanistic basis for the hallmark deficits of AD.

Beyond medial temporal interactions, VBM analysis further identified the precentral gyrus and hippocampus axis. Although the precentral gyrus is traditionally considered a motor hub, recent evidence suggests its early cortical thinning in AD and predictive value for MCI-to-AD conversion[36]. Convergent findings also demonstrate reduced functional coupling between the precentral gyrus and hippocampus in AD, with connectivity strength paralleling MMSE scores[37]. In earlier disease stages, including pMCI vs. sMCI and NC vs. EMCI comparisons (Figure S6B and C), VBM revealed strong paracentral lobule (PCL)–precentral gyrus (PreCG) interactions. Despite their canonical involvement in distinct motor and visuospatial processes, these regions display reduced connectivity and metabolic coupling in AD[38,39]. Their coordinated decline may contribute to atypical AD manifestations such as visuospatial dysfunction and motor coordination impairments, underscoring the heterogeneity of network-level vulnerability.

Finally, the cerebellar midline pair Vermis7 and Vermis8 showed high model-derived importance (Figure S6D). These adjacent regions participate in affective and self-referential processing, and altered vermal connectivity has been observed in children with ADHD[40]. Aberrant coactivation of Vermis7 and Vermis8 may underlie emotional dysregulation and impaired task modulation in ADHD. The convergence of our model-derived interactions with prior evidence of vermal dysconnectivity underscores the biological plausibility of our framework and highlights cerebellar pairwise markers as promising candidates for transdiagnostic biomarker development.

S2.3.4 Consistent multimodal biomarkers revealed by region-level ablation

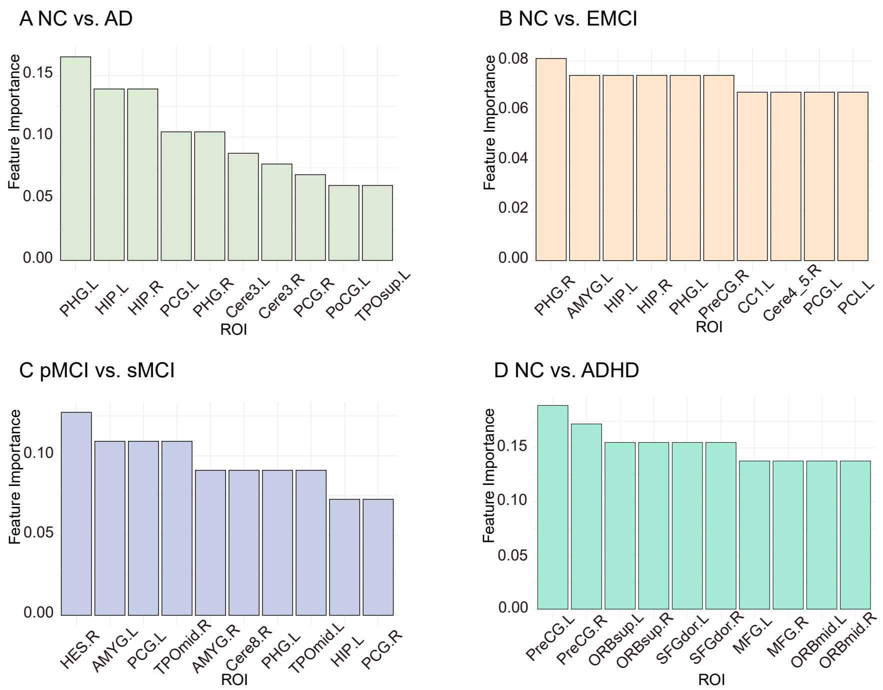

Figure S7: Region-level ablation analysis cross imaging modalities for (A) NC vs. AD, (B) NC vs. EMCI, (C) pMCI vs.sMCI, and (D) NC vs. ADHD tasks.

To evaluate whether CAMM can uncover stable and biologically meaningful biomarkers, we performed region-level ablation across all modalities, in which the same brain region was simultaneously removed from all the modalities. This design allows us to probe multimodal contributions at the regional level and to assess whether cross-modal integration highlights consistent patterns beyond any single modality. Figure S7 summarizes the most influential regions identified for each classification task.

We first examined the most established contrast, NC vs. AD (Figure S7A). Here, the parahippocampal gyrus and hippocampus emerged as the dominant contributors, reaffirming their well-known role as the earliest and most vulnerable loci in AD pathology[34,35]. Beyond these medial temporal structures, the posterior cingulate gyrus, postcentral gyrus, superior temporal pole, and cerebellum (lobule III) also ranked highly, suggesting that sensorimotor and temporal association cortices, together with cerebellar circuits, provide additional discriminative signals at the dementia stage[41]. Building on this foundation, the NC vs. EMCI task (Figure S7B) revealed a broader network already perturbed at the prodromal stage. While the parahippocampal gyrus and hippocampus again appeared among the most important features, the amygdala and precentral gyrus also rose in prominence, together with contributions from the cingulate cortex, paracentral lobule, and cerebellum (lobule IV–V). These findings indicate that distributed cortical–subcortical alterations emerge early, expanding beyond medial temporal lobe pathology[41]. Progression within the MCI spectrum was further illuminated by the pMCI vs. sMCI comparison (Figure S7C). The highest-ranked regions included the Heschl gyrus, amygdala, parahippocampal gyrus, hippocampus, and temporal poles (middle subdivision). The involvement of auditory, affective, and semantic-memory circuits highlights subtle but functionally diverse disruptions that help stratify progression risk among MCI patients[42–44]. Finally, when shifting from neurodegeneration to a neurodevelopmental disorder, NC vs. ADHD (Figure S7D) implicated a distinct frontal–motor network. The precentral gyrus, superior frontal gyrus (dorsal subdivision), middle frontal gyrus, and orbitofrontal cortices (superior and middle subdivisions) were the most discriminative, in line with established evidence of impaired motor control and executive regulation in ADHD[45–47].

These results demonstrate a coherent progression: from medial temporal vulnerability in AD, through distributed limbic and sensorimotor involvement in EMCI, to subtle multimodal disruptions marking MCI progression, and finally to distinct frontal and motor alterations in ADHD. Across conditions, certain regions, including the hippocampus, parahippocampal gyrus, posterior cingulate cortex, amygdala, temporal poles, and precentral gyrus, emerge as stable multimodal biomarkers. This hierarchical view underscores how cross-modal integration can disentangle core pathological substrates from modality-specific noise and provide biologically meaningful, interpretable markers for brain disorder prediction.

S2.3.5 Inter-modality dependency revealed by region-level bimodal ablation

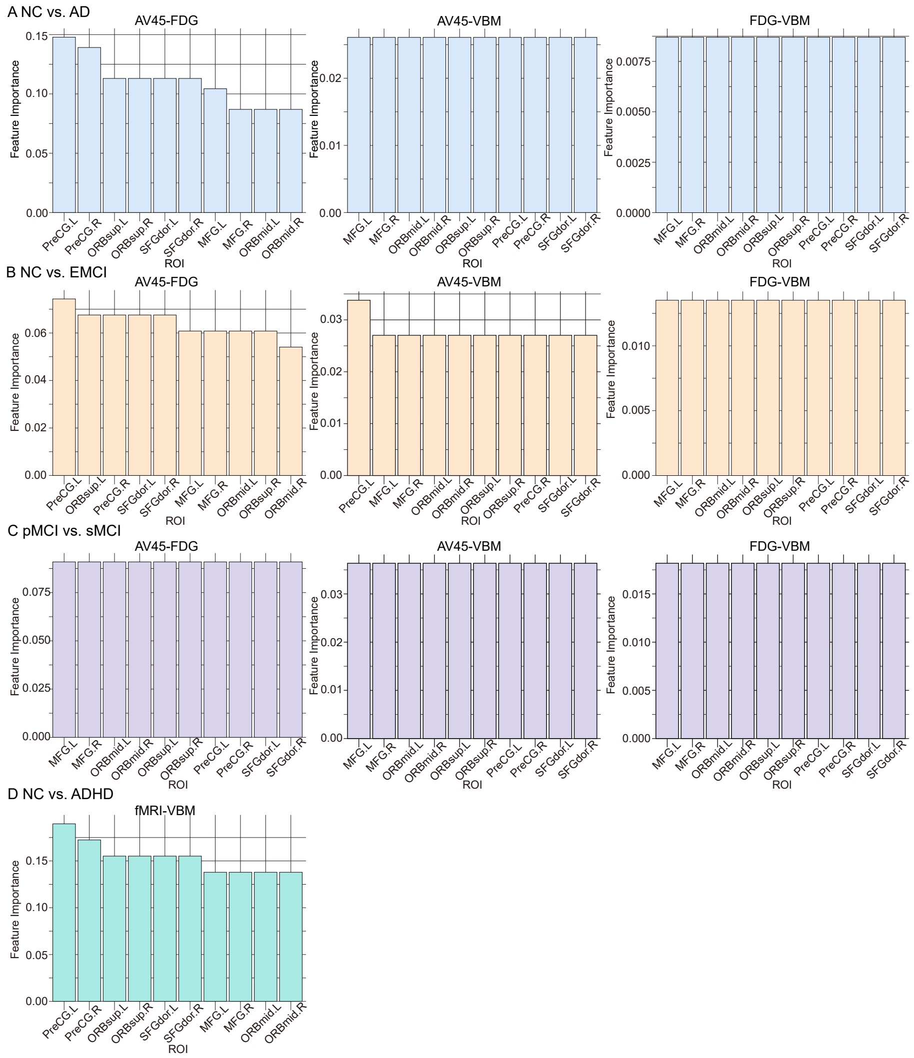

Figure S8: Region-level bimodal ablation analysis for inter-modality dependency, showing top-ranked regions for (A) NC vs. AD, (B) NC vs. EMCI, (C) pMCI vs. sMCI, and (D) NC vs. ADHD under different modality pairs.

To explore inter-modality dependencies and identify complementary biomarkers across imaging domains, we conducted region-level bimodal ablation experiments in which the same brain region was jointly removed from two modalities. As shown in Figure S6, we evaluated three bimodal combinations—AV45-FDG, AV45-VBM, and FDG-VBM—for AD-related classification tasks, and fMRI-VBM for the NC vs. ADHD task. The goal is to assess how the importance of a region is preserved or enhanced across modalities, thereby revealing synergistic regions whose cross-modal representations contribute jointly to disease prediction[48,49].

In the NC vs. AD task (Figure S8A), the precentral gyrus, dorsal superior frontal gyrus, and superior orbitofrontal cortex emerge as top-ranking regions specifically in the AV45–FDG combination. These frontal areas are not traditionally emphasized in AD studies, yet their prominence in the metabolic–amyloid context underscores their potential involvement in attentional control and motor planning—functions often subtly impaired in early AD[50]. Notably, the orbitofrontal cortex (particularly the superior subdivision), which plays a role in reward processing and executive control, may reflect early compensatory mechanisms detectable through amyloid and metabolic coupling. By contrast, in combinations involving VBM, feature importances appear more uniformly distributed, suggesting that structural signals alone may dilute regional distinctiveness and highlight the value of molecular context in capturing disease-relevant patterns. Moving to earlier disease stages (Figure S8B and C), frontal involvement again surfaced selectively in the AV45–FDG pairing for NC vs. EMCI, with the middle frontal gyrus, precentral gyrus, and orbitofrontal subdivisions emerging as salient. However, such specificity diminished in AV45–VBM and FDG–VBM, and disappeared entirely in pMCI vs. sMCI across all modality pairs. This progression indicates that the discriminative power of frontal regions is concentrated in amyloid–metabolic coupling, while structural features remain too diffuse to highlight early-stage changes[51]. Beyond AD, the NC vs. ADHD task (Figure S8D) with fMRI–VBM provided a complementary perspective. Here, the precentral gyrus bilaterally was most prominent, followed by the dorsal superior frontal gyrus and superior orbitofrontal cortex, with middle orbitofrontal regions showing weaker contributions. This hierarchy aligns with prior evidence linking ADHD to deficits in motor control and executive regulation[45]. Because these regions become salient only when functional and structural signals are combined, the result highlights convergent dysfunction across modalities rather than modality-specific effects.

The bimodal ablation results demonstrate that certain regions retain predictive power only under joint cross-modal evaluation. Such robustness underscores inter-modality complementarity, wherein structural, metabolic, and functional features integrate to capture disease-specific pathophysiology[48]. Importantly, regions such as the precentral gyrus, frontal cortices, and orbitofrontal areas emerge as functional bridges that sustain predictive relevance in both neurodegenerative and neurodevelopmental contexts.

S2.4 Pathway enrichment analysis of genes used for T-RRI construction

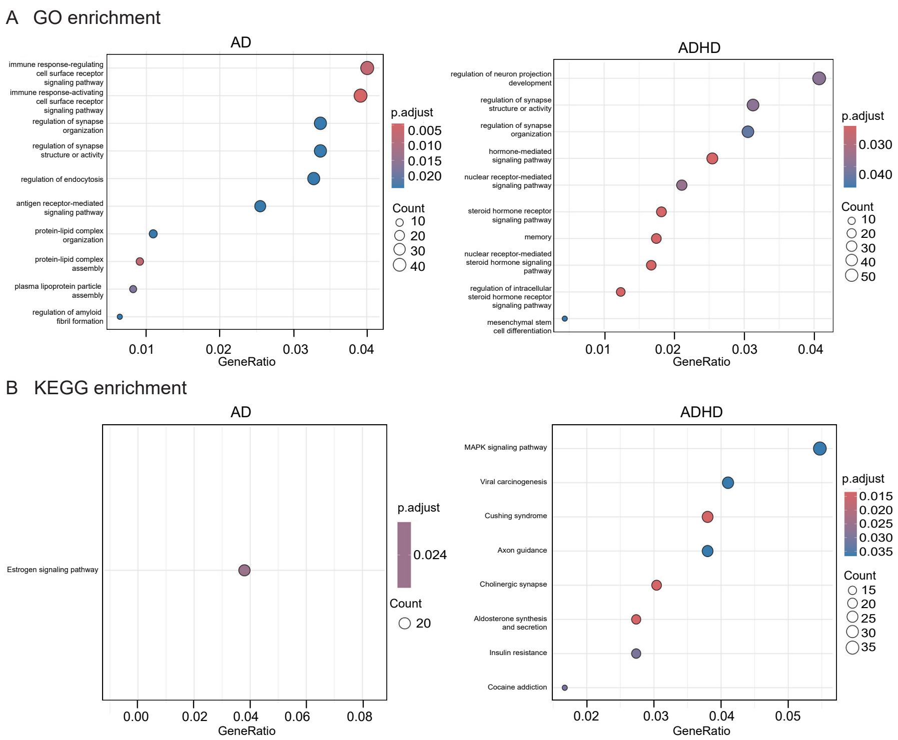

Figure S9: Pathway enrichment analysis of disease-associated genes used for T-RRI construction. (A) GO enrichment results for Alzheimer's disease (AD) and attention-deficit/hyperactivity disorder (ADHD). (B) KEGG pathway enrichment results for AD and ADHD.

To examine whether the genes used for T-RRI construction are associated with biologically meaningful and disease-relevant molecular mechanisms, we performed GO and KEGG pathway enrichment analyses for disease-associated genes and showed the results in Figure S9.

For AD, GO enrichment analysis showed that AD-associated genes were predominantly enriched in immune-related signaling processes, particularly receptor-mediated mechanisms involved in immune activation and regulation. This enrichment pattern highlights the central role of microglial immune responses in AD pathogenesis and is consistent with accumulating evidence linking dysregulated immune receptor signaling to neuroinflammation and amyloid-related pathology[52,53]. Complementary KEGG pathway analysis further revealed significant enrichment in hormone-dependent signaling pathways, with estrogen-mediated signaling emerging as a key component. Given the established role of estrogen signaling in modulating synaptic function, neuroinflammation, and cognitive resilience, these results suggest that impaired hormone-related neuroprotective mechanisms may increase vulnerability to AD[54,55]. Together, the GO and KEGG results indicate that the AD-associated genes used in our framework capture molecular processes closely linked to AD progression.

For ADHD, GO and KEGG enrichment analyses revealed that ADHD-associated genes were primarily involved in neurodevelopmental processes, synaptic regulation, and signal transduction. Enrichment patterns consistently pointed to dysregulation of neuronal projection development and synaptic structure or activity, supporting the view of ADHD as a neurodevelopmental disorder characterized by altered neural circuit formation and synaptic function[56,57]. In addition to core neurodevelopmental mechanisms, enrichment results also implicated signaling and immune-related processes that may influence early brain development. Dysregulated intracellular signaling cascades related to neuronal growth, differentiation, and synaptic plasticity, together with immune activation pathways associated with inflammatory or viral responses, suggest that both intrinsic developmental programs and immune-related perturbations may contribute to ADHD vulnerability[58,59]. Collectively, these findings demonstrate that the genes used for T-RRI construction are enriched in disease-relevant pathways for both AD and ADHD, providing biological support for the transcriptomic priors employed in our framework.

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